

$(\sigma E[x])$	$\longrightarrow (\sigma E[v])$ where $v = \text{lookup}[\sigma, x]$	[var]
$(\sigma E[(\text{if } \#false\ e_1\ e_2)])$	$\longrightarrow (\sigma E[e_2])$	[if-false]
$(\sigma E[(\text{if } v\ e_1\ e_2)])$	$\longrightarrow (\sigma E[e_1])$ where $(\text{not } (\text{equal? } \#false\ v))$	[if-true]
$(\sigma_0 E[(\text{set! } x\ v)])$	$\longrightarrow (\sigma_I E[(\text{void})])$ where $\sigma_I = \text{ext1}[\sigma_0, (x\ v)]$	[set!]
$(\sigma E[(\text{fix } v_{rec}\ v_{arg}\ \dots)])$	$\longrightarrow (\sigma E[(v_{rec}\ (\text{fix } v_{rec}))\ v_{arg}\ \dots])$ where $(\text{lambda } (x\ \dots)\ e) = v_{rec}$	[fix]
$(\sigma E[(\text{lambda } (x\ \dots_1)\ e)\ v\ \dots_1])$	$\longrightarrow (\sigma_I E[e])$ where $\sigma_I = \text{ext}[\sigma_0, (x\ v), \dots]$	[app]
$(\sigma_0 E[(\text{let } ([x\ v]\ \dots)\ e_{body})])$	$\longrightarrow (\sigma_I E[e_{body}])$ where $\sigma_I = \text{ext}[\sigma_0, (x\ v), \dots]$	[let]
$(\sigma E[(\text{begin } v_0\ v_I\ v_{ns}\ \dots)])$	$\longrightarrow (\sigma E[(\text{begin } v_I\ v_{ns}\ \dots)])$	[begin]
$(\sigma E[(\text{begin } v)])$	$\longrightarrow (\sigma E[v])$	[begin0]
$(\sigma E[(\text{slot } x_{field}\ v_{struct})])$	$\longrightarrow (\sigma E[v])$ where $v = \text{lookup}[\sigma, ((x_s\ v_s)\ \dots), x_{field}]$	[slot]
$(\sigma_0 \text{ prog})$	$\longrightarrow (\sigma_I E[(\text{ptr } x)])$ where $\sigma_I = \text{ext1}[\sigma_0, (x\ v)], \text{prog} = E[(\text{box } v)]$	[box]
$(\sigma E[(\text{unbox } v)])$	$\longrightarrow (\sigma E[v_{unboxed}])$	[unbox]
$(\sigma_0 E[(\text{set-box! } v_{ptr}\ v_{new})])$	$\longrightarrow (\sigma_I E[(\text{void})])$ where $\sigma_I = \text{ext1}[\sigma_0, (x_{addr}\ v_{new})]$	[set-box!]
$(\sigma E[(+ \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } +\ (\text{number } \dots))])$	[add]
$(\sigma E[(- \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } -\ (\text{number } \dots))])$	[subtract]
$(\sigma E[(\text{num->string } \text{ number})])$	$\longrightarrow (\sigma E[(\text{number->string } \text{ number})])$	[num->string]
$(\sigma E[(= \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } =\ (\text{number } \dots))])$	[num=]
$(\sigma E[(< \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } <\ (\text{number } \dots))])$	[num<]
$(\sigma E[(> \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } >\ (\text{number } \dots))])$	[num>]
$(\sigma E[(\leq \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } \leq\ (\text{number } \dots))])$	[num<=]
$(\sigma E[(\geq \text{ number } \dots)])$	$\longrightarrow (\sigma E[(\text{apply } \geq\ (\text{number } \dots))])$	[num>=]
$(\sigma E[(\text{append string } \dots)])$	$\longrightarrow (\sigma E[(\text{apply string-append } (\text{string } \dots))])$	[append]